

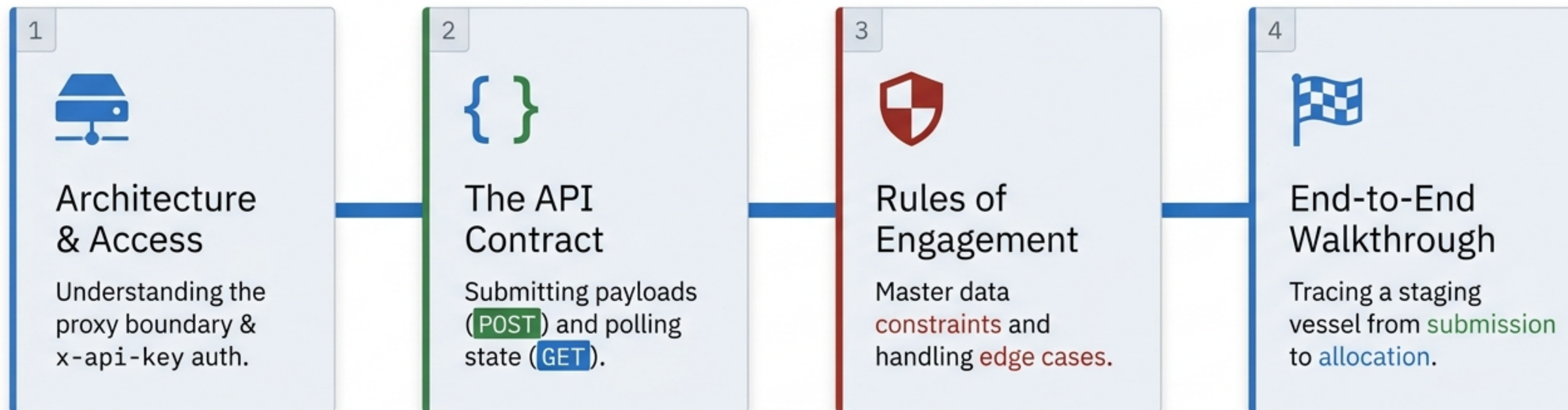
SHIPPING INSTRUCTION API

Integration Playbook for the Jetty Planning System (JPS)

```
~$ connect jps --env staging
```

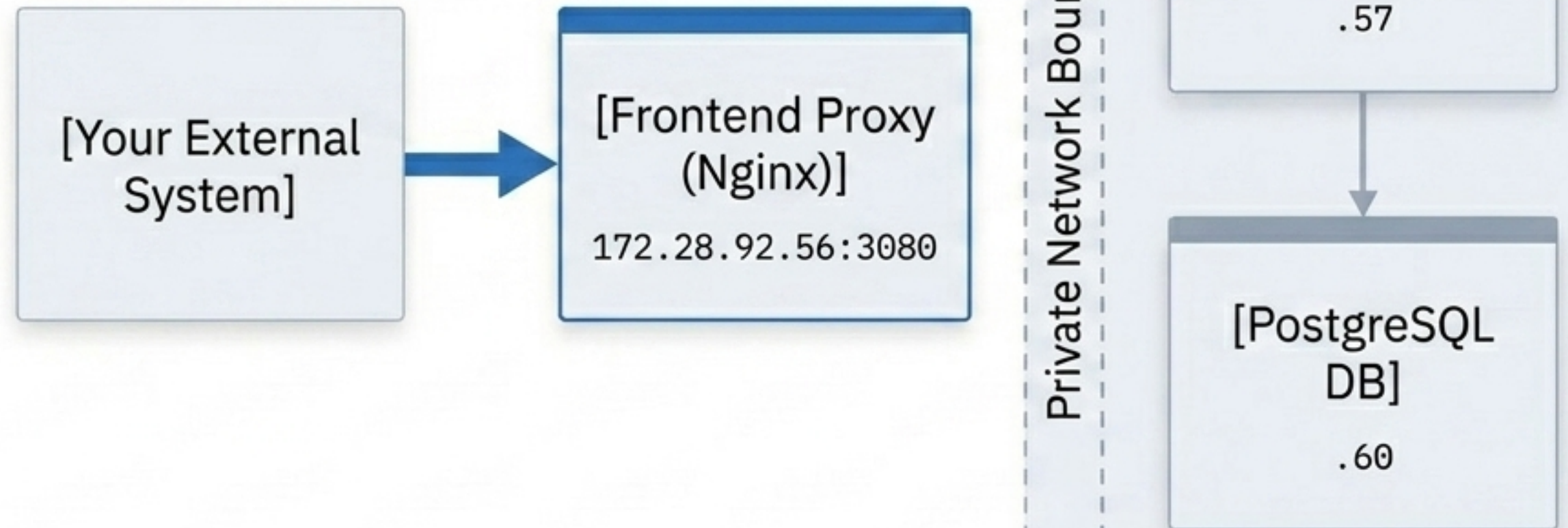
STAGING: v3.0

INTEGRATION ROADMAP



THE INTEGRATION BOUNDARY

- **Asynchronous Workflow:**
Push shipping instructions via API; JPS operators manually handle approval and jetty allocation in their UI.
- ✓ **Clear Boundaries:**
Implement a simple POST-and-poll mechanism. No webhook receivers or complex OAuth flows required.
- 🌀 **Staging Isolation:**
External developers must target the staging proxy. Direct backend access is blocked.



AUTHENTICATION & ACCESS

- **Single Header Validation:**
All endpoints are secured by a static key injected via the x-api-key HTTP header.
- **Key Generation Process:**
 - **External App Owner:**
Requests a key.
 - **Jetty Planning System Owner:**
Generates and securely provides the static key.
- **Scoped Permissions:** Your key is strictly bound to your partner identity and allowed ports.
 - Currently, only Port ID 1 (BONTANG) is ready. Future ports will be added later.
- **Zero Overhead:** No token signing. Store the key securely in your .env vault.

request.js

```
const headers = {  
  'Content-Type': 'application/json',  
  'x-api-key': process.env.JPS_API_KEY // e.g., jps_live_a73fc...  
};
```


SUBMITTING PAYLOADS (POST)

- 🕒 **Idempotency is Key:**
external_reference guarantees uniqueness per API key. Safe to retry on timeouts; duplicates return a 409 Conflict.
- 📦 **Flat Structure:** Combine vessel data, ETA, and cargo details into a single JSON schema.
- ✅ **Instant Acknowledgment:**
Success immediately
Success immediately yields a 201 Created with a JPS ID, defaulting to a Pending state.

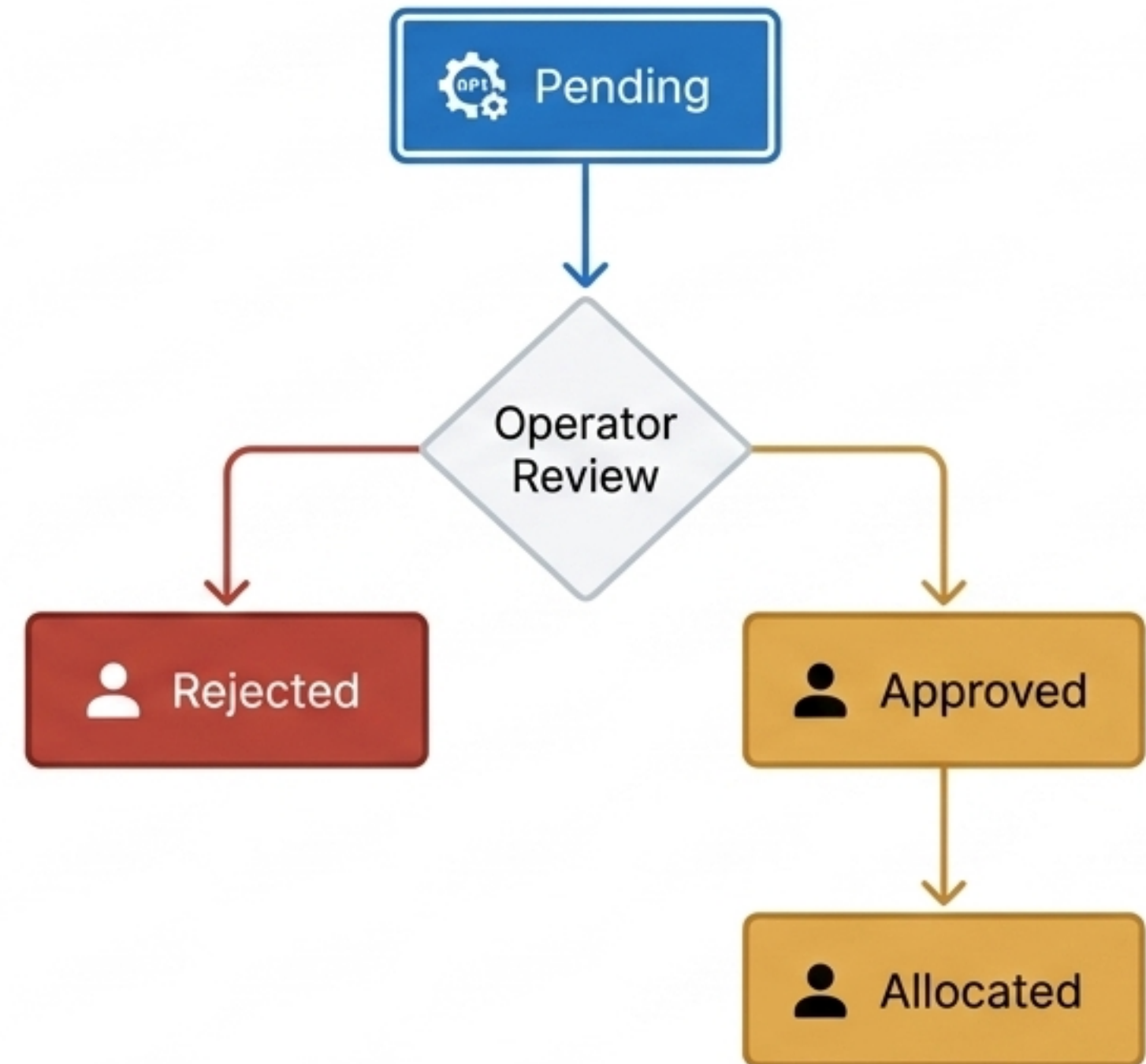
request.json

```
{
  'external_reference': 'EOS-EXPORT-2026-091', // Idempotency key
  'port_id': 1,
  'purpose': 'Loading',
  'cargo': [
    {
      'cargo_type': 'CP0',
      'tonnage': 5000,
      'unit': 'MT'
    }
  ]
}
```


TRACKING THE LIFECYCLE (GET)

- ↔ **Dual Lookup:** Retrieve live status using either the JPS ID or your external_reference.
- 👤 **Human-in-the-Loop:** Transitions out of Pending are asynchronous actions executed by JPS operators in the web UI.
- 🕒 **Gentle Cadence:** Poll at a maximum rate of once every 5 minutes until reaching an end-state.

```
GET /shipping-instructions?  
external_reference=EOS-EXPORT-2026-091
```



MASTER DATA STRICTNESS

Do This / Not That

	REJECTED (Full Name)		REQUIRED (JPS short_name)
✗	cargo_type: 'CRUDE PALM OIL'	✓	cargo_type: 'CP0'
	REJECTED (Unmapped)		REQUIRED (Mapped Equivalent)
✗	cargo_type: 'PK0'	✓	cargo_type: 'CPK0'

Short Codes Only

The payload must exactly match the internal JPS short_name.

Category Constraints

All cargo lines in one instruction must share the same physical state (Solid or Liquid).

Dynamic Validation

Sending an unknown cargo type triggers a 400 error containing a real-time list of valid short codes.

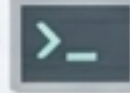
DIAGNOSTICS & EDGE CASES

HTTP Code	Internal Trigger	Required Action
400 VALIDATION_ERROR	Bad payload (e.g., unknown cargo)	Do Not Retry - Fix the JSON.
401 / 403	Invalid Key or Forbidden Port	Do Not Retry - Check .env configuration.
409 DUPLICATE_REFERENCE	Same external ref submitted	Recover - Use GET to retrieve existing ID.
429 / 500	Rate Limited (>120/min) or Server Error	Smart Retry - Use exponential backoff.

Tracing Support: Always log the request_id from the error envelope. It is vital for tracing issues with the JPS Backend Team.

CASE EXAMPLE: THE STAGING VESSEL

System App



T+0

POST external_reference: SI-TEST-001.
Receives 201 Created (ID: 41, Status: Pending).

T+5m

GET status.
Status: Still Pending.

T+10m

GET status.
Status: Receives 200 OK (Status: Approved).

T+15m

GET status.
Status: Receives 200 OK (Status: Allocated,
Jetty: Jetty 2). Workflow complete.

JPS Operator



T+8m

Logs into staging UI, verifies vessel details,
clicks Approve.

T+12m

Opens Allocation module, assigns to Jetty 2.

STAGING TO PRODUCTION

- [✓] Health check passing on 172.28.92.56:3080
- [✓] Staging key stored securely (Port 1 validated)
- [✓] Cargo payload strictly mapped to JPS short_names
- [✓] Idempotency tested via duplicate POSTs (409 Conflict handled)
- [✓] Asynchronous polling loop established (max 1 request / 5m)

Code the integration. Run the tests. Request production keys.